

Induced potential, transformers and the National Grid

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 13 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	State what is induced by the generator effect.	A potential difference (and a current, in a complete circuit).	☐
2 ★	State what is needed for the generator effect to occur.	A conductor moving relative to a magnetic field, or a changing magnetic field around a conductor.	☐
3 ★	Give three ways to increase the size of the induced potential difference.	Use a stronger magnetic field, move faster, or use more turns on the coil.	☐
4	Describe how to reverse the direction of the induced potential difference.	Reverse the direction of movement or of the magnetic field.	☐
5 ★	State the difference between an alternator and a dynamo.	An alternator generates alternating current; a dynamo generates direct current.	☐
6	Name the effect used in microphones.	The generator effect	☐
7 ★	Describe how a microphone works.	Sound makes a diaphragm and coil vibrate in a magnetic field, inducing a changing potential difference.	☐
8	Name the effect used in transformers.	The generator effect	☐
9 ★	Describe the structure of a transformer.	A primary coil and a secondary coil wound on an iron core.	☐
10	Describe what a step-up transformer does.	It increases the potential difference.	☐
11	Describe what a step-down transformer does.	It decreases the potential difference.	☐
12 ★	Compare the numbers of turns on the coils of a step-up transformer.	More turns on the secondary coil than on the primary.	☐

QUESTION		ANSWER	✓
13	Compare the numbers of turns on the coils of a step-down transformer.	Fewer turns on the secondary coil than on the primary.	☐
14 ★	State how power input and power output are related in an ideal transformer.	They are equal.	☐
15	Name the type of current supplied to the primary coil of a transformer.	Alternating current	☐
16	Name the type of current induced in the secondary coil of a transformer.	Alternating current	☐
17 ★	Explain why a transformer needs an iron core.	Iron is easily magnetised, so it carries the changing magnetic field from the primary coil to the secondary coil.	☐
18 ★	State what the magnetic field of an induced current does.	It opposes the change that caused it.	☐
19 ★	Explain why a transformer does not work with a direct current.	Direct current gives a constant magnetic field; induction needs a changing field.	☐
20	State the equation linking the potential differences and numbers of turns of a transformer.	p.d. across primary ÷ p.d. across secondary = turns on primary ÷ turns on secondary ($V_p \div V_s = n_p \div n_s$)	☐
21	State the equation for a transformer that is 100% efficient.	p.d. × current in the secondary = p.d. × current in the primary ($V_s \times I_s = V_p \times I_p$)	☐
22 ★	Describe the potential difference–time graph for an alternator.	It alternates: a wave going equally above and below zero.	☐
23 ★	Describe the potential difference–time graph for a dynamo.	It rises and falls but never goes below zero.	☐